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12

ONE-DIMENSIONAL STEADY FLOWS

12.1 General program for one-dimensional flows

All calculations in this chapter are made with either the 'general one-dimensional' program DSMC1.FOR, or with several variants of this program that have special adaptions for the study of shock waves. The term 'one-dimensional' is taken here to mean that there can be macroscopic flow gradients in only one direction. There can be finite flow velocity components in more than one direction and, in addition to the plane flow geometry, the program can be applied to flows with cylindrical or spherical symmetry. The geometry is specified by setting the parameter IFX to 0 for plane flows, to 1 for cylindrical flows, or to 2 for spherical flows. All DSMC calculations are unsteady but, since this chapter deals only with steady flows, the sampling is concerned primarily with the steady flow that is established at large times. The gas is a simple gas or a gas mixture of monatomic or diatomic gases, and the DSMC collision procedures follow those of program DSMC0R.FOR.

The macroscopic flow gradients must be in the x-direction and, for cylindrical and spherical flows, the axis or centre must be at x=0 so that x becomes the radius. There is an 'inner' and an 'outer' boundary at the fixed values of x that are set by the variables XB(1) and XB(2). Each of these flow boundaries is specified through the variables IB(1) and IB(2) as one of:

(i) The axis of a cylindrical flow or the centre of a spherical flow.

(ii) A plane of symmetry for IFX=0, or a specularly reflecting cylinder or sphere for IFX=1 or 2, respectively.

(iii) A solid diffusely reflecting surface with complete accommodation to a specified temperature. For plane or cylindrical flows, this surface may have a velocity in the y-direction. The axis of the cylindrical flow is along the z-axis, so that this is a circumferential velocity.

(iv) A boundary with an external stream of specified composition, density, and temperature. The stream may also have a velocity component in the x-direction.

(v) An interface with a vacuum.

The initial state of the flowfield may be either a vacuum or a uniform and macroscopically stationary gas of specified composition, density, and temperature. A number of the subroutines are common to the earlier programs and the full program is listed on the enclosed disk.

The flowfield is divided into $n_c \equiv$ MNC cells, and each of the cells is divided into NSC sub-cells. The cells of uniform width that were used in the programs for homogeneous gas studies become inefficient for problems that involve large changes in density. Should cells of non-uniform width be used, it is important that there should be an efficient procedure for determining the appropriate cell for a molecule at a given location x. An analytical expression for the cell number as a function of x is available if the cell widths are in arithmetic or geometric progression. The geometric progression is generally better suited to the density variations that occur in real flows and is defined by

$$a + ar + ar^2 + \ldots + ar^{n-1} = a(1-r^n)/(1-r).$$

The molecule location is normalized to lie between 0 and 1 by setting

$$\hat{x} = (x - x_1)/(x_2 - x_1),$$

where x_1 and x_2 are the boundaries of the region. The variation in cell width is defined by the ratio $c_w \equiv$ CWR of the width of the cell adjacent to the boundary at x_2 to the width of the cell adjacent to x_1. The multiplier r is therefore given by

$$r = c_w^{1/(n_c-1)}.$$

Also, since the normalized width of the flow region is unity, the sum of the series must also be unity, and the width of the first cell is

$$a = (1-r)/(1-r^{n_c}).$$

The normalized coordinate of the lower valued boundary of cell m is

$$a(1-r^{m-1})/(1-r),$$

and the cell number is given by equating the normalized coordinate to this expression, solving for m, and truncating the solution to the nearest integer. The final result is, therefore,

$$m = \lfloor 1 + \ln\{1-(1-r)\hat{x}/a\}/\ln(r) \rfloor. \tag{12.1}$$

Each time a molecule is moved, the boundaries of the original cell are checked to determine whether the cell has changed and, if it has, eqn (12.1) is used to determine the number of the new cell. The sub-cells in a given cell are of uniform width.

The flow can have a uniform 'gravitational' acceleration g along the x-axis. This is applied through the elementary equations

$$\Delta x = u\Delta T + \tfrac{1}{2}g(\Delta t)^2$$

and

$$\Delta u = g\Delta t.$$

The logic for the interaction of molecules with solid boundaries has not been extended to include the effects of this acceleration and the option should be used only with type (iv) and type (v) boundaries in plane flows.

The more general boundaries that are employed in this program require a number of new procedures. The first is for the generation of typical molecules that are diffusely reflected from a surface. The distribution function for these molecules is identical to that for the molecules crossing a boundary from an external stationary equilibrium gas. For a surface or boundary normal to the x-axis, the distribution function for the velocity component normal to the boundary follows from eqns (4.20) and (4.1) as

$$f_u \propto u \exp(-\beta^2 u^2).$$

The normalization condition is that the integration of this distribution with respect to u from 0 to ∞ must be equal to unity. Therefore,

$$f_u = 2\beta^2 u \exp(-\beta^2 u^2), \tag{12.2}$$

and the distribution function for $\beta^2 u^2$ follows as

$$f_{\beta^2 u^2} = \exp(-\beta^2 u^2).$$

This distribution is identical with that of eqn (C11) so that, from eqn (C12), a typical value of u is given by

$$u = \{-\ln(R_f)\}^{1/2}/\beta. \tag{12.3}$$

Eqn (12.3) is implemented for the reflected molecules in the subroutine REFLECT1. The distribution function for a velocity component parallel to the boundary is the same as that for a velocity component within a stationary gas. Both parallel components of a diffusely reflected molecule are generated by a single call of the subroutine RVELC.

Should there be a stream velocity component u_0, the thermal velocity component is $u' = u - u_0$, and the distribution function for the ratio of the thermal velocity component to the most probable thermal speed is

$$f_{\beta u'} \propto (\beta u' + s_n) \exp(-\beta^2 u'^2), \tag{12.4}$$

where $s_n = \beta u_0$ is the speed ratio based on the normal component of the stream velocity. The acceptance–rejection method is required for the generation of typical velocity components from this distribution. The maximum of this distribution occurs at

$$\beta u' = \{(s_n^2 + 2)^{1/2} - s_n\}/2$$

and the ratio of the probability to the maximum probability is

$$\frac{P}{P_{max}} = \frac{2(\beta u' + s_n)}{s_n + (s_n^2 + 2)^{1/2}} \exp\left(\frac{1}{2} + \frac{s_n}{2}\{s_n - (s_n^2 + 2)^{1/2}\} - \beta^2 u'^2_n\right). \tag{12.5}$$

The procedure generates a value of $\beta u'$ that is uniformly distributed over a range that extends well into the tails of the distribution and covers virtually all entering molecules, but is subject to the condition that $\beta u' + s_n$ must be directed into the flow. The value of P/P_{max} is then evaluated and is compared with the next random fraction R_f. β is the inverse of the most probable molecular speed in the entering gas.

The parallel components of velocity of the entering molecules are again generated by a call of the subroutine RVELC. The rotational energy of the stream, reflected, and entering molecules are generated by calls of the subroutine SROT. This subroutine includes the procedure that was the subject of subroutine SIE in DSMC0V.FOR and DSMC0D.FOR. The number flux of the entering molecules is calculated directly from eqn (4.22). This involves the error function, and the function subroutine ERF has been included. The cross-sectional area of plane flows has been assumed to be unity. The cylindrical flow has an axial length of unity and the cross-sectional area at radius r is πr^2. The cross-sectional area of a spherical flow at radius r is taken to be πr^2. Molecules that leave the flow across either a stream or vacuum boundary are removed form the simulation. This is achieved in the subroutine REMOVE by simply replacing this molecule by molecule NM, where NM is the total number of molecules. NM is then reduced to NM–1.

Advantage is taken of cylindrical or spherical symmetry in that the radius is the only spatial coordinate that must be stored for each molecule, and the results are obtained as a function of the radius. At the same time, the logic associated with the three-dimensional motion of the molecules must be exact. It is assumed that, at the beginning of each time step, the molecule lies in the x-axis. The y-velocity component and, in the case of spherical symmetry, the z-velocity component cause the molecule to move off the axis. However, the new radius may be calculated, and this position coordinate is stored for the molecule at the end of the time step. However, this involves a rotation of the direction of the molecule, and the velocity components of the molecules must be transformed accordingly. Before the transformations, the new position on the x-axis is

$$x' = x_i + u\Delta t\,, \tag{12.6}$$

where x_i is the initial radius of the molecule, u is the velocity component in the x-direction, and Δt is the time step. The molecule moves off the axis through a distance

$$d_n = v\Delta t$$

in a cylindrical flow, or

$$d_n = \{(v\Delta t)^2 + (w\Delta t)^2\}^{1/2}$$

in a spherical flow. The new radius is given by

$$x = (x'^2 + d_n^2)^{1/2} \tag{12.7}$$

while the sine and cosine of the rotation angle are

$$\sin\theta = d_n/x$$

and (12.8)

$$\cos\theta = x'/x\,.$$

The transformed velocity component in the radial (i.e. the x) direction is

$$u' = u\cos\theta + u_c\cos\theta, \tag{12.9}$$

where the circumferential velocity component u_c is v in a cylindrical flow or $(v^2+w^2)^{1/2}$ in a spherical flow. The transformed value of the circumferential velocity component is

$$u'_c = -u\sin\theta + u_c\cos\theta. \tag{12.10}$$

This becomes the new y-velocity component v in a cylindrical flow while, in a spherical flow, an azimuth angle ϕ may be chosen at random between 0 and 2π to give

$$v' = u_r\sin\phi \quad \text{and} \quad w' = u_r\cos\phi. \tag{12.11}$$

These equations are implemented in the subroutine AIFX. This is called at the end of the trajectory element or when the molecule intersects with a boundary.

The calculation of the point of intersection with a boundary must be based on three-dimensional geometry. Eqns (7.85)–(7.87) may be applied to each boundary in turn to calculate the ratio s_d of the distance along the molecular trajectory to any intersection with the boundary to the distance d that is moved during the time step. The direction cosines are given by

$$l_1 = u\Delta t/d, \quad m_1 = v\Delta t/d, \quad \text{and} \quad n_1 = w\Delta t/d,$$

where

$$d = \{(u\Delta t)^2 + (v\Delta t)^2 + (\varepsilon-1)(w\Delta t)^2\}^{1/2}.$$

The z-direction cosine applies only to spherical flows, and the parameter ε is 1 for cylindrical flows and 2 for spherical flows. The coefficients of x^2, y^2 and, for spherical flows, z^2 are unity, and the coefficient a_{44} is $-r^2$, where r is the radius of the cylinder or sphere. All other coefficients are zero, as are y_i and z_i. The ratio s_d is therefore given by the quadratic

$$\tfrac{1}{2}s_d^2 + (u\Delta t x_i/d^2)s_d + \tfrac{1}{2}(x_i^2 - r^2)/d^2 = 0. \tag{12.12}$$

The molecule collides with the surface if there is a real solution between 0 and 1. There can be two real solutions for the inner boundary, and the lesser solution then gives the point of intersection. Note that there can be no intersection with the inner boundary unless the value of x' from eqn (12.6) is less than r. Similarly, there can be no intersection with the outer boundary unless the final radius x is greater than r. The fraction s_d is evaluated in the subroutine RBC.

The surface properties of any diffusely reflecting solid surface are sampled as the reflections occur in the routine REFLECT1. The number flux, pressure, shear stress, and heat transfer to these surfaces are output in the subroutine OUT1. The surface data distinguishes between the contributions that are made by the incident and reflected molecules. The heat transfer also records separately the contributions from the translational and rotational modes.

12.2 The viscosity coefficient of argon

The first application of the program DSMC1.FOR is essentially a numerical experiment for the measurement of the coefficient of viscosity in argon. Plane Couette flow is the flow between two plane surfaces, one of which is stationary and the other is moving with a velocity in the plane of the surface. This is generated here by setting the inner boundary as a stationary diffusely reflecting surface and the outer boundary as a diffusely reflecting surface at the same temperature, but with a finite velocity in the y-direction. The gas is initially set to the same temperature as the walls, but it may be either stationary or have a uniform stream velocity gradient.

The surfaces are maintained at a temperature of 273 K with a spacing of 1 m, and the gas is argon at a number density of 1.4×10^{20}. The VHS molecular model is employed with the molecular diameter as listed in Table A2 and the mean free path λ is, from eqn (5.69), 0.00925 m. The Knudsen number based in the surface spacing is, therefore, 0.00925. The velocity of the outer surface is 300 ms^{-1}, which corresponds to a speed ratio of 0.89. The velocity profile is shown in Fig. 12.1. The velocity gradient is very nearly uniform although there is a barely perceptible decrease in the gradient within about 10 to 20 mean free paths from the surfaces. There is a velocity slip of just over 3 ms^{-1} at each surface and this is, as expected, very close to $\lambda\, dv/dx$.

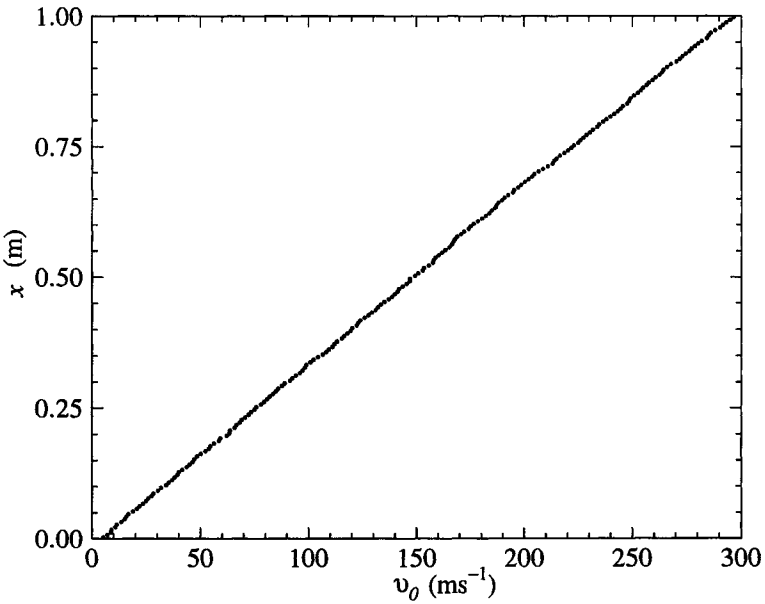


Fig. 12.1 The velocity profile in the Couette flow at $(Kn) = 0.00925$.

The incompressible continuum solution would be a constant pressure and constant temperature gas with a uniform velocity gradient, and no velocity slip at the surface. The coefficient of viscosity of argon at 273 K (on which the molecular diameter was based) is 2.117×10^{-5} Nsm^{-2}, and the shear stress on the surfaces would be 6.35×10^{-3} Nm^{-2}. The incompressible solution is, however, incompatible with the energy conservation principle. The motion of the wall against the shear stress would generate an energy input to the gas of 1.905 Wm^{-2}, and this would raise the temperature of the gas until the heat that was transferred to the surfaces equalled the energy input. The pressure must be uniform through the gas, and the temperature non-uniformities must lead to density variations that invalidate the incompressibility assumption.

The DSMC solution exhibits the expected compressibility effects. The initial value of the pressure is 0.5277 Nm^{-2} and, because of the heating due to the viscous dissipation, this increases to a steady state value of 0.549 Nm^{-2}. The pressure is uniform across the flow, while the temperature rise in the central portion of the flow is approximately 15 K, or just over 5% of the surface temperature. The DSMC temperature profile is shown in Fig. 12.2. The temperature slip at the surfaces is approximately 0.6 K, and this is close to $\lambda\, dT/dx$. Because of the uniform pressure, the density change must be the inverse of the temperature change, and the density profile is shown in Fig. 12.3.

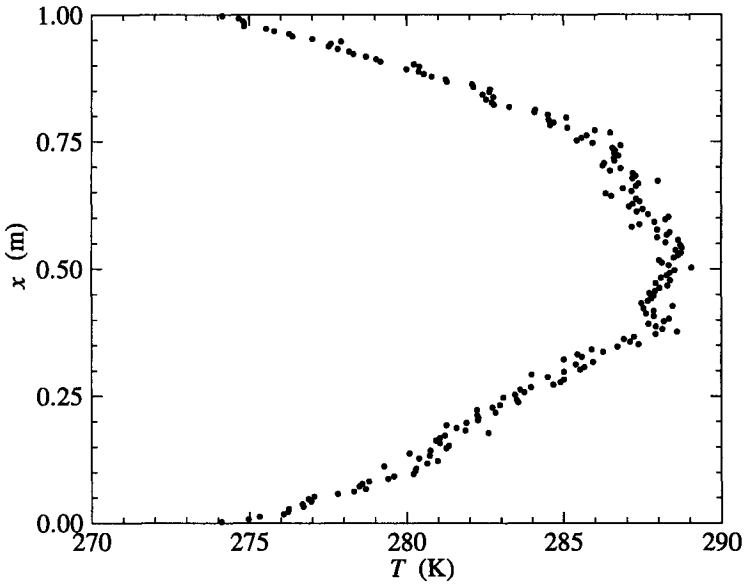


Fig. 12.2 The temperature profile in the Couette flow.

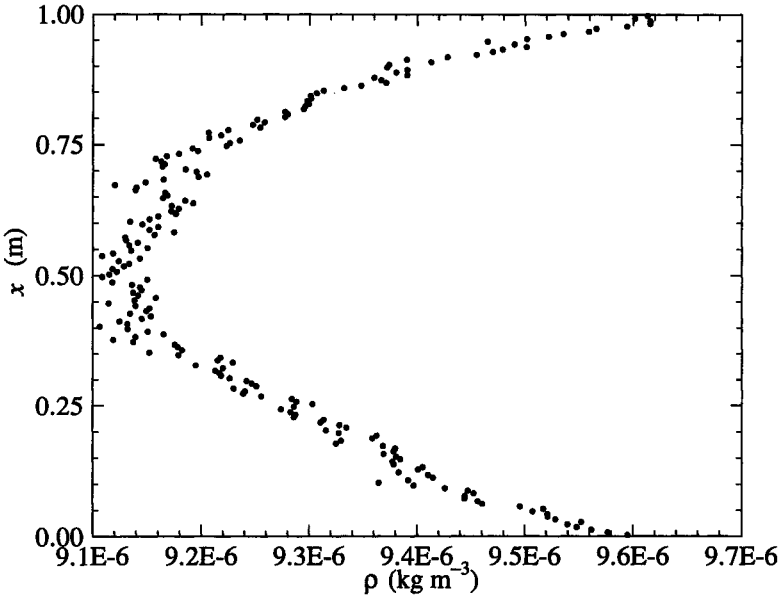


Fig. 12.3 The density profile in the Couette flow.

These results are based on time-averaged sampling over the interval from 0.1 seconds to 0.375 seconds. This run was made with the listed data in the subroutine DATA1, and employs the option that sets a uniform velocity gradient in the initial gas. Runs were also made for a uniform initial gas and steady flow was established at about 0.5 sec. The test case involved the calculation of 7.7×10^6 collision events and 1.5×10^9 molecular moves. Almost 270,000 molecules collided with each surface and the standard deviation associated with the surface pressure is 0.2%. The sample in each cell was more than 1.3×10^6, and the expected standard deviation in the macroscopic properties is less than 0.1%. However, the flow is subsonic and both the shear stress and heat flux are essentially based on small differences between large numbers. The shear stress is 1.1% of the pressure, and the net heat flux to the surfaces is less than 1% of the incident energy flux. This means that the scatter of a single number can be as high as 10% but, in addition to the sampling of the shear stress as a surface property, it may be sampled as a macroscopic flow property in each cell. The boundary conditions are such that some applications involve a stream velocity component in the x-direction and others a component in the y-direction, but none involve a finite component in each direction. The shear stress for this case can therefore be written from eqn (1.52) as

$$\tau_{xy} = -n\,\overline{m\,u\,v}\,. \tag{12.13}$$

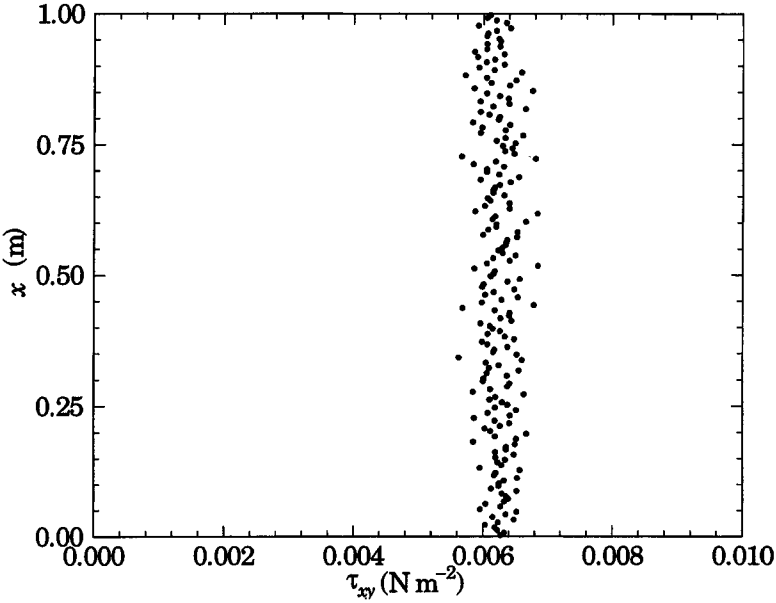


Fig. 12.4 The shear stress profile in the Couette flow.

Similarly, since the stream velocity must consist only of u_0 or v_0, the heat flux vector in the x-direction can be written from eqn (1.53) as

$$q_x = n(\tfrac{1}{2}\overline{mc^2u} - \overline{mu^2}u_0 + \overline{m}u_0^3 - \overline{muv}v_0 - \tfrac{1}{2}\overline{mc^2}u_0 + \overline{\varepsilon u} - \bar{\varepsilon}u_0). \quad (12.14)$$

The shear stress profile is shown in Fig. 12.4 and, while there is significant statistical scatter, it is constant across the flow. The average value of the stress over the flow was 0.00624 Nm^{-2} and the velocity gradient was 293 (m/s)/m. These values correspond to a coefficient of viscosity of 2.13×10^{-5} Nsm^{-2}, which may be compared with the nominal value of 2.117×10^{-5} Nsm^{-2} at a temperature of 273 K. The average temperature across the flow is 283 K and, with a temperature–viscosity index of 0.81, the nominal coefficient is 2.18×10^{-5} Nsm^{-2}. Given the scatter associated with the shear stress and the variation of temperature across the flow, the agreement is satisfactory, although the discrepancy is rather larger than would be expected. The mean separation between collision partners was 1.7×10^{-4} m which is very small in comparison with the mean free path, and the other computational parameters were satisfied by similarly wide margins.

The work done by the moving surface at the average shear stress is 1.87 Wm^{-2}, while the sampled net heat transfer to the inner and outer surfaces is 0.85 Wm^{-2} and 1.03 Wm^{-2}, respectively. The heat flux vector sampled from eqn (12.14) is shown in Fig. 12.5. The scatter in q_x is very high, and better results are obtained here if u_0 is forced to its steady flow value of zero.

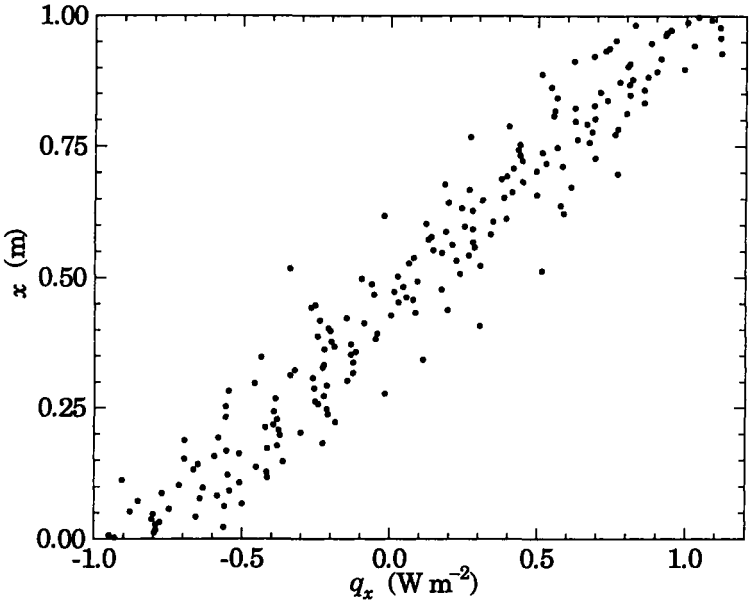


Fig. 12.5 The heat flux vector in the Couette flow.

Because the statistical scatter is a fraction of the undisturbed flow quantities, accurate results are more easily obtained for large disturbance flows than for small disturbance flows. The calculation was therefore repeated for a surface velocity of 1000 m/s, which corresponds to a speed ratio of 2.67. The Mach number based on the wall speed and the temperature of the undisturbed gas is 2.93, and much of the flow is supersonic. The velocity gradient and the shear stress increase by a factor of just over three, but the temperature gradient and heat flux vector increase by a factor of more than ten. The net heat flux to a surface increases from less than one percent of either the incident or reflected flux to more than ten percent of these quantities. The resulting effect on the scatter is quite dramatic, as can be seen from a comparison of Figs. 12.5 and 12.6.

The high speed velocity profile of Fig. 12.7 is qualitatively different from the relatively low speed profile of Fig. 12.1. The velocity gradient in Fig. 12.1 is linear to within the statistical scatter, while the high speed velocity profile is nonlinear over the complete flowfield. This nonlinearity is due to the temperature dependence if the coefficient of viscosity. The temperature profile remains parabolic in form, and the maximum temperature is 438 K. The coefficient of viscosity at the centre of the flow is therefore about 40% higher than that near the surfaces and, in order for the shear stress to remain constant across the flow, the velocity gradient in the cooler gas near the surfaces must be correspondingly larger than that near the centre.

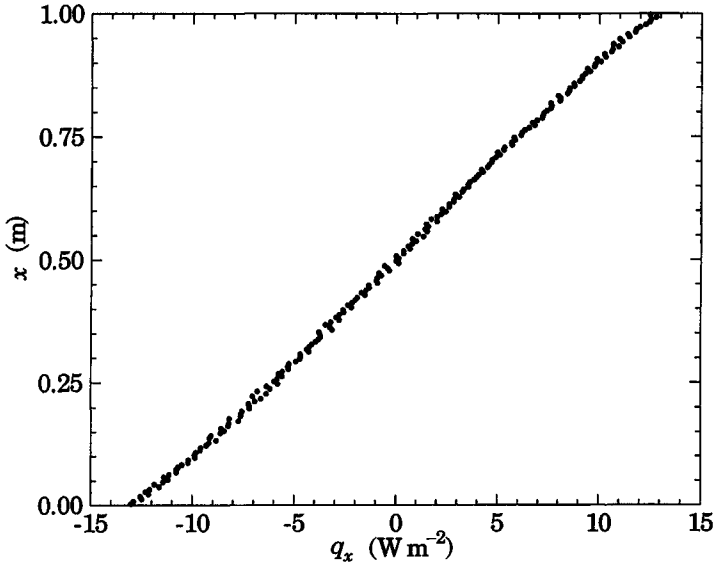


Fig. 12.6 The heat flux in the high-speed Couette flow.

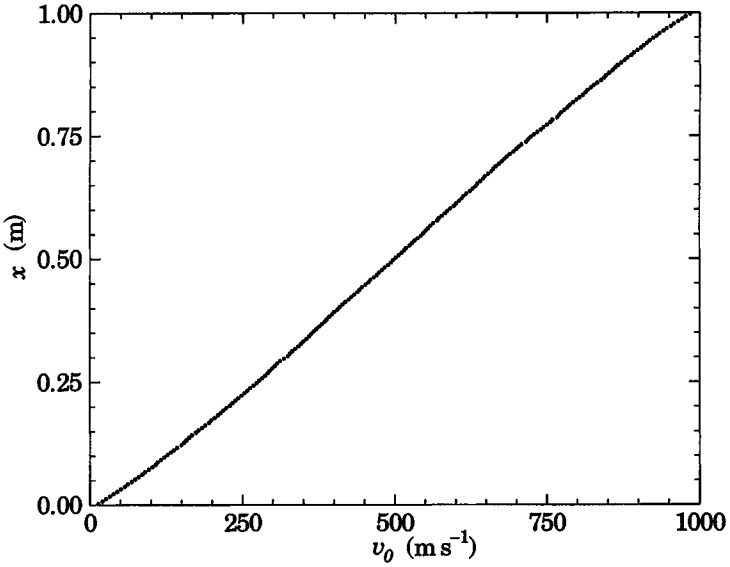


Fig. 12.7 The velocity profile in the high-speed Couette flow.

Rather than use an average velocity gradient and an average temperature as was done for the low-speed case, it is preferable to directly sample the coefficient of viscosity at the reference temperature μ_{ref} from the results in the file DSMC1.OUT. The required equation is

$$\mu_{ref} = \tau_{xy} (\Delta x / \Delta v_0) (T_{ref}/T)^{\omega}. \tag{12.15}$$

The quantities τ_{xy}, x, v_0, and T are output for each cell and the numerical array may be edited out as a separate file and read into a post-processing program for the evaluation of the transport properties. The program that was used here based Δx and Δu_0 for a particular cell on the differences between the values in the two adjacent cells. The coefficient of viscosity was then obtained by averaging over all cells except those in the Knudsen layers very close to the surfaces. The resulting coefficient of viscosity at the reference temperature of 273 K was 2.10×10^{-5} Nsm^{-2}, and this compares well with the value of 2.117×10^{-5} Nsm^{-2} on which the VHS molecular diameter was based.

The velocity and temperature 'slips' at the surfaces were again very close to the product of the local mean free path with the velocity gradient and the temperature gradient, respectively.

The thermal effects in the high-speed case are sufficiently large to allow the determination of the coefficient of heat conduction in addition to the coefficient of viscosity. The equation for the evaluation of the reference value of this coefficient is

$$K_{ref} = q_x (\Delta x / \Delta T) (T_{ref}/T)^{\omega}. \tag{12.16}$$

and, since q_y also appears in DSMC1.OUT, values at each cell are readily calculated. The results are hopelessly erratic in the central portion of the flow, but the averaged value in the outer 30% of the flow was 0.0166 $Wm^{-1}K^{-1}$. This is in reasonable agreement with the measured value of 0.0164 $Wm^{-1}K^{-1}$ that is quoted by Chapman and Cowling (1970) for argon at 273 K. The resulting Prandtl number is 0.666 which is in excellent agreement with the theoretical value of 2/3 for a monatomic gas.

The temperature correction in the high-speed case is very significant and the calculation was repeated for a surface temperature of 180 K so that the average temperature was much nearer to 273 K. The resulting viscosity coefficient was 2.11 Nsm^{-2}, and the heat transfer coefficient was 0.0167 $Wm^{-1}K^{-1}$.

The derivations from and comparisons with theory have been based on the results from the first approximation of the Chapman–Enskog theory, as discussed in §3.5. The second approximation, that is introduced by an additional term in the Sonine polynomial, would increase the theoretical coefficient of viscosity in argon by less than one half of one percent. This is of the order of the statistical scatter in the DSMC results. In addition, the Burnett terms that result from the inclusion of the next term in the Chapman–Enskog expansion itself are probably of the same magnitude. These terms would include the so-called 'thermal stresses' that arise from the contribution of the temperature gradient to the stress tensor.

12.3 The viscosity of an argon–helium mixture

The viscosity of an argon–helium mixture at 293 K has been discussed in detail by Chapman and Cowling (1970). The coefficient of viscosity of argon is 13.6% greater than that of helium but, as long as the percentage of helium is less than about 83%, the coefficient of viscosity of the mixture exceeds that of argon. The maximum viscosity coefficient occurs for about 60% helium and is 4% higher than the coefficient for pure argon. In order to check that the DSMC method reproduces this behaviour, calculations similar to those in the previous section were made for an argon–helium mixture. One difference is that the VHS model is replaced by the VSS model. Also the parameter ISPD has been set to unity, and this requires the effective diameter and viscosity for the argon–helium collisions to be set explicitly in the data rather than being set to the mean values.

The measured values of diffusion coefficient may be used with the VSS scattering parameters that are listed in Appendix A to define the effective diameter $(d_{12})_{\text{ref}}$ of the cross-collisions at a given reference temperature. This procedure is analogous to that used in §4.3 to define a molecular diameter from the coefficient of viscosity, and the result is

$$(d_{12})_{\text{ref}} = \left(\frac{3(\alpha_{12}+1)\{2kT_{\text{ref}}/(\pi m_r)\}^{\frac{1}{2}}}{8(5-2\omega_{12})\, n\, (D_{12})_{\text{ref}}} \right)^{\frac{1}{2}}. \qquad (12.17)$$

The diameter is that at the mean value of $c_r^{2\omega-1}$ in an equilibrium gas at the reference temperature. It should again be emphasized that the use of an equilibrium quantity to define the reference diameter is a matter of convenience and has no physical significance. Alternatively, the diameter may be set to the mean diameter of eqn (1.35), and the values of the diffusion coefficient used to calculate the appropriate value of the VSS scattering parameter. This is the preferred approach and is adopted in Appendix A. Both approaches were used here and, although the scattering parameters differ, the viscosity coefficients were indistinguishable.

Since the 'overshoot' in the coefficient of viscosity is only 4%, the level of statistical scatter in the DSMC result is of critical importance. The flow is sampled at regular intervals in order to build up a cumulative sample. However, the sample in each cell changes slowly as a consequence of diffusion and it is not easy to choose the minimum time interval that leads to independent samples. In addition, the net shear stress is the small difference between large quantities and the size of the cumulative sample is not a reliable guide to the expected accuracy. The best indication of the accuracy of the DSMC result is given by the time history of the cumulative sample. This history for the coefficient of viscosity in pure argon at 293 K in a gas of VSS molecules is plotted in Fig. 12.8 for more than 3000 individual samples. The sampling started at 0.08 s which was the assumed time for the establishment of steady flow. The sample size at 0.6 s was therefore 25 times larger than that at 0.1 s, and the statistical scatter would be expected to decline by a factor of five over the plotted interval.

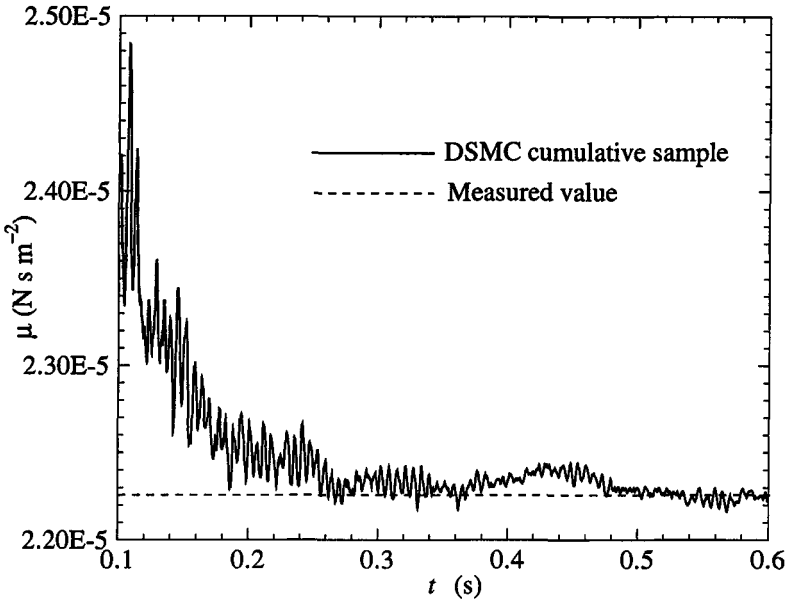


Fig. 12.8 The coefficient of viscosity in argon at 293 K.

The progressive reduction in the level of the statistical scatter is in line with expectations, and the DSMC result is consistent with the measured value. Most of the computations for the gas mixture involved a sample to approximately 0.4 seconds. Fig. 12.8 indicates that the standard deviation of the statistical scatter should be less than 1%. This means that most results should be within 1% of the correct value, but extreme cases could be 2% or more in error. The results from the DSMC calculations with program DSMC1.FOR are shown in Table 12.1. These are in agreement with the measured values to within the expected statistical scatter.

Table 12.1 Viscosity coefficients in He-Ar mixtures at 293 K.

Helium (%)	Measured μ (Nsm^{-2})	Calculated μ (Nsm^{-2})
0.0	2.2275×10^{-5}	2.22×10^{-5}
19.9	2.2707×10^{-5}	2.24×10^{-5}
37.1	2.3095×10^{-5}	2.31×10^{-5}
63.4	2.3161×10^{-5}	2.27×10^{-5}
100.0	1.9604×10^{-5}	1.99×10^{-5}

12.4 The Prandtl number of nitrogen

As shown by eqn (12.14) the internal energy ε of the molecules contributes to the heat flux vector. This leads to an increase in the heat flux, but the specific heat also changes, and the degree of departure of the Prandtl number from the monatomic gas value of 2/3 is of interest.

The high speed Couette flow calculation was therefore repeated with the VHS model for nitrogen. The post-processing program was again used to produce averaged values of the viscosity and heat transfer coefficients at 273 K. The viscosity coefficient was found to be 1.645×10^{-5} N s/m^2, while the VHS diameter was based on the value of 1.656×10^{-5} N s/m^2 that is quoted by Chapman and Cowling (1970). The corresponding heat transfer coefficient was 0.0232 W/(m K). This may be compared with the measured value of 0.0240 W/(mK) from Chapman and Cowling who note that the estimated error in the experiments that led to these values is not less than one percent. The DSMC and measured values of the Prandtl number are then 0.74 and 0.72, respectively. The Prandtl number of diatomic gases generally approximates 3/4 rather than the 2/3 for monatomic gases.

12.5 The self-diffusion coefficient of argon

Experiments to measure the self-diffusion coefficient generally employ different isotopes in order that the otherwise identical molecules may be distinguished. The fact that the isotopes have slightly different masses adds to the experimental error. The 'numerical experiment' with the DSMC method is facilitated by the fact that identical molecules may be designated as both species 1 and species 2. The VSS model for argon was employed with the molecular properties as listed in Tables A1 and A3. The data for program DSMC1.FOR was set with both boundaries as 'stream boundaries' (IB(1)=IB(2)=4) with a stationary gas as the stream. The stream at the inner boundary at $x = 0$ was comprised entirely of species 1 and that at the outer boundary at $x = 1$ m was comprised entirely of species 2. In this case, the streams were at a number density of 1.4×20 m^{-3} and a temperature of 273 K. The flowfield was plane with the locations of the boundaries, the number of cells, and the number of sub-cells were the same as in the listed data for DSMC1.FOR. The initial composition of the gas was equal parts of the (physically identical) species 1 and species 2.

After the gas had come to a steady state, there was a linear variation of composition across the flowfield, as shown in Fig. 12.9. The molecules that cross the stream boundary from the gas are discarded and there is a 'composition slip' at the boundaries. The gas is macroscopically uniform and stationary but, as shown in Fig. 12.10, the diffusion velocities are quite substantial. It should be noted that the diffusion coefficients are inversely proportional to the number density, so that the diffusion velocities in a gas at standard temperature and pressure would be smaller than those shown here by a factor of approximately 200,000.

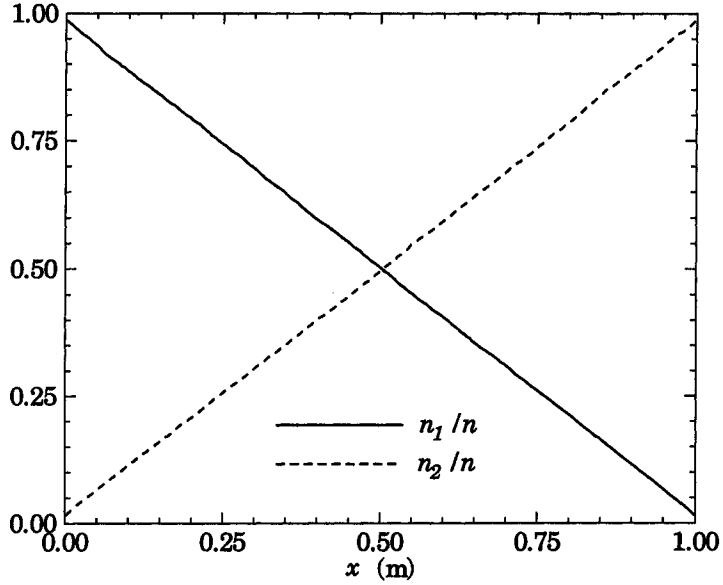


Fig. 12.9 The concentration gradients in the self-diffusion problem.

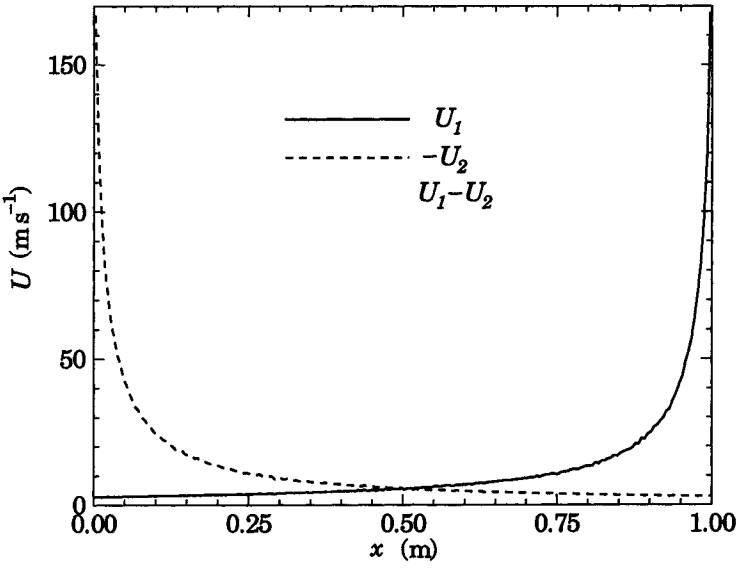


Fig. 12.10 The diffusion velocities in the self-diffusion problem.

For this problem, eqn (3.91) enables the self-diffusion coefficient to be written

$$D_{11} = D_{12} = -(U_1 - U_2)\frac{n_1 n_2}{n^2}\frac{\Delta x}{\Delta (n_1/n)}. \qquad (12.18)$$

The concentration gradient is uniform across the flow and the variation of the diffusion velocities is such that, as expected, D_{11} is also constant across the flow to within the expected statistical scatter. The averaged value of D_{11} at a temperature of 273 K and a number density of 1.4×10^{20} m^{-3} was found to be 2.95 m^{2}s^{-1}. This value corresponds to 1.55×10^{-5} at S.T.P. and is in good agreement with the measured value of 1.57×10^{-5} m^{2}s^{-1} (Chapman and Cowling 1970).

The calculation was repeated for the VHS model ($\alpha = 1$) and also for the value of $\alpha = 1.67$ proposed by Koura and Matsumoto (1991). The resulting S.T.P. values for D_{11} were 1.26×10^{-5} m^{2}s^{-1} and 1.72×10^{-5} m^{2}s^{-1}, respectively. As expected from eqn (3.76) D_{11} is proportional to $\alpha + 1$, and the value of α given in Table A3 is the only value that leads to the correct coefficient of self-diffusion.

12.6 Mass diffusion in an argon–helium mixture

In order to test the values of the VSS scattering parameter in Table A3, the 'diffusion cell' calculation was repeated with an argon reservoir at the inner boundary and a helium reservoir at the outer boundary. Because of the small molecular diameter of helium, the average mean free path is larger than that in the pure argon calculation. The number densities in the reservoirs was therefore doubled to 2.8×10^{20} in order to keep the local Knudsen numbers well in the range of values for which the Chapman–Enskog theory should be valid.

The species concentration and diffusion velocity profiles are shown in Figs 12.11 and 12.12. These may be compared with Figs 12.9 and 12.10 for the argon self-diffusion problem. The symmetry disappears and the point at which the species concentrations are equal moves from the midpoint of the flow to a location nearer to the argon reservoir. The diffusion velocity in the helium is larger than that in the argon by a factor that approximates the square root of the mass ratio. There is a qualitative difference in the velocity profiles in that there is a shallow minimum in the helium diffusion velocity.

Eqn 12.8 can again be used to determine the diffusion coefficient. Even though there are very large gradients near the boundaries, this was again near uniform across the flow. The averaged S.T.P. value of D_{12} in the central 80% of the flow was 6.39×10^{-5} m^2 s^{-1}, and that across the complete flow was 6.33×10^{-5} m^2 s^{-1}. The measured S.T.P. value of D_{12} in an argon–helium mixture is given by Chapman and Cowling as 6.41×10^{-5} m^2 s^{-1}. The expected error in both the experiments and the DSMC calculation is of the order of 1%.

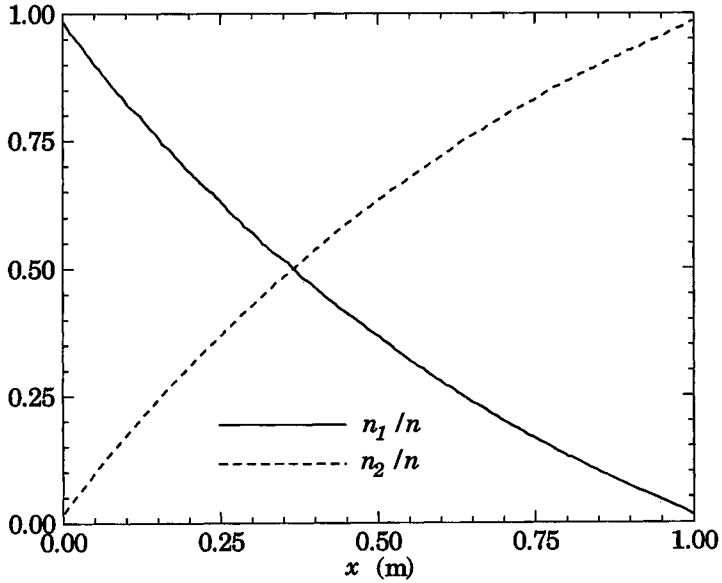


Fig. 12.11 The concentration gradients in Ar–He diffusion.

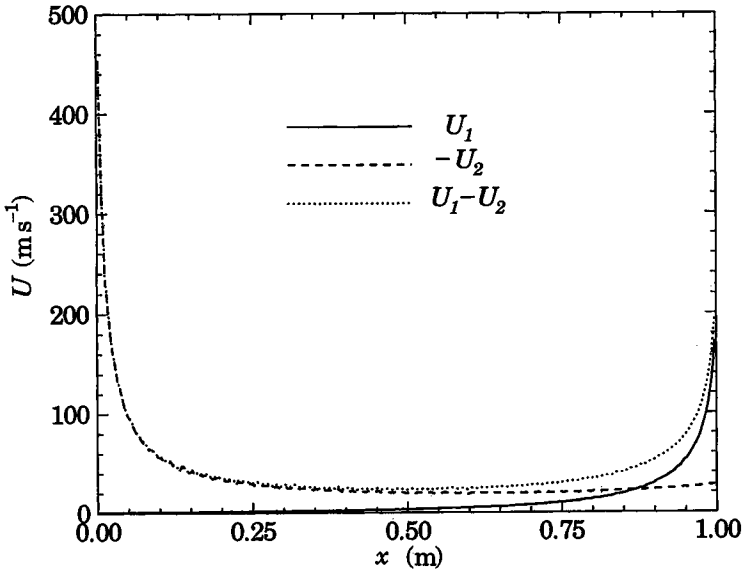


Fig. 12.12 The diffusion velocities in Ar–He diffusion.

12.7 Thermal diffusion

The composition of the gas in the high-speed Couette flow of the 37.1% helium, 62.9% argon mixture is shown in Fig. 12.13. Even though the boundary conditions do not induce mass diffusion, and the flow is at a uniform pressure, there is a partial separation of the two species with a reduction in the concentration of argon at the centre of the flow. This is due to thermal diffusion which, as discussed in §3.5, causes the larger and generally heavier molecules to become concentrated in regions of low temperature.

Thermal diffusion is best studied quantitatively in a simpler flow with no mass velocities. The data in the program DSMC1.FOR may be set such that the inner boundary is a diffusely reflecting boundary at 300 K and the outer surface is a stream boundary with a mixture of equal parts (by number) of helium and argon at a temperature of 2000 K and a number density of 1×10^{20}. The gas is initially set to the stream conditions and, during the unsteady phase of the flow, the gas is cooled at the surface and the consequent increase in the density induces a flow of gas towards the surface. The gas eventually becomes stationary at the stream pressure, but with a strong temperature gradient that produces separation effects due to thermal diffusion.

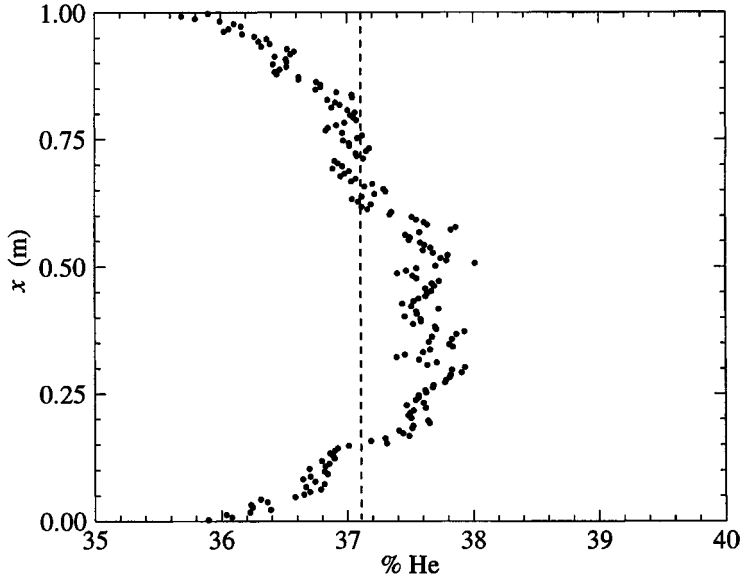


Fig. 12.13 Separation effects in the high-speed Couette flow for a 37.1% He–Ar mixture.

The steady state does not involve any diffusion velocities and eqn (3.91) reduces to

$$\mathrm{d}(n_1/n)/\mathrm{d}x = k_T \,\mathrm{d}\ln(T)/\mathrm{d}x\,. \tag{12.19}$$

The thermal diffusion ratio k_T depends strongly on the composition, and it is preferable to quantify the effect through the thermal diffusion factor which is defined by

$$\alpha_T = \{n^2/(n_1 n_2)\}\, k_T\,. \tag{12.20}$$

Eqn (12.19) can then be integrated to give

$$\ln(n_1/n_2) = -\alpha_T \ln(T) + C, \tag{12.21}$$

where C is a constant.

The temperature and species concentration profiles in the steady state are shown in Figs 12.14 and 12.15. The mean free path in the 'stream' at $x = 1$ m is approximately 0.02 m and, since the density is here inversely proportional to the temperature, the mean free path at the surface is less than 0.003 m. The local Knudsen numbers are therefore well within the limits of validity of the Chapman-Enskog theory and the only effects of the rarefaction are the temperature slips at the boundaries. The separation is substantial and extends over the whole simulated region.

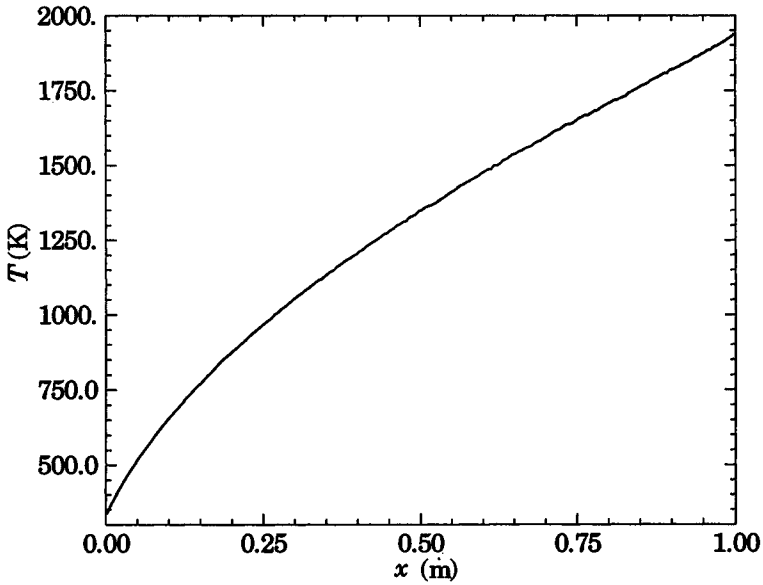


Fig. 12.14 The temperature profile between a surface at 300 K and a stationary gas of equal parts by number of He and Ar at 2000 K.

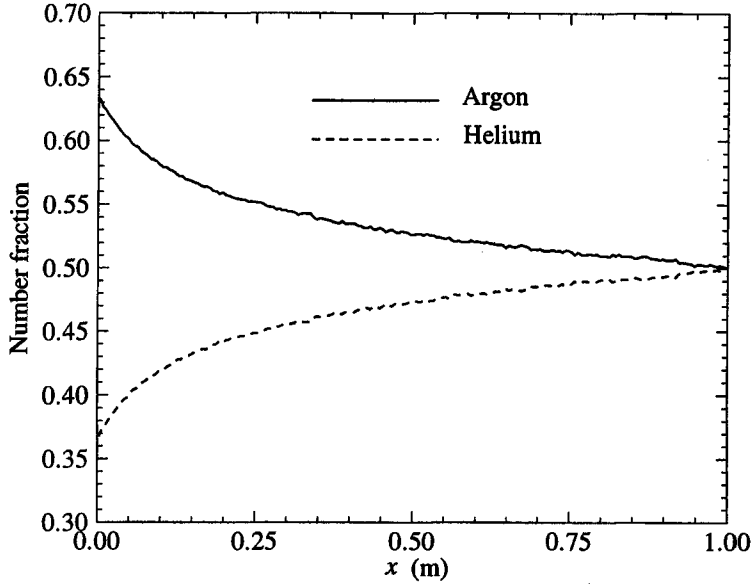


Fig. 12.15 The separation due to thermal diffusion.

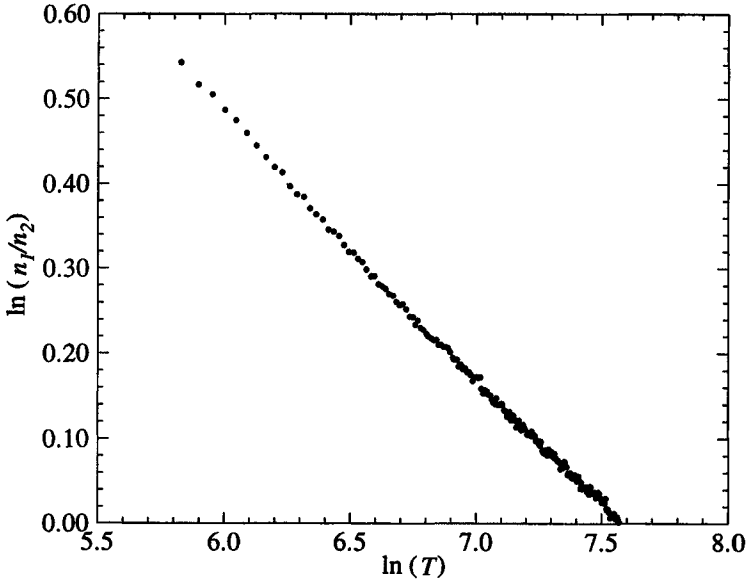


Fig. 12.16 Plot for α_T as the slope (see eqn (12.21)).

Fig. 12.16 plots the logarithmic variables that appear in eqn (12.21) in such a way that the slope should be negative and its absolute value equal to the thermal diffusion factor. There is a slight systematic reduction in the slope at high temperatures, but the slope at the lower temperatures of the order of 300 K gives a value of 0.35 for α_T in a He–Ar mixture. The experimental value of this quantity that is given by Chapman and Cowling (1970) is 0.38.

Thermal diffusion is caused by the variations in size of the molecules. The calculation was repeated with ω=1 for all species and with the reference diameters chosen such that all species had the same collision cross-section. For ω=1, the cross-section is inversely proportional to the relative speed, so that all collision pairs are equally probable. This case is analogous to Maxwell molecules, and it is known that thermal diffusion does not occur in a Maxwell gas. As expected, there was no separation in this case.

12.8 The diffusion thermo-effect

This is the inverse effect to thermal diffusion and is a heat flux in a gas at uniform temperature, but with an inhomogeneity in composition. This effect should have been be present in the He–Ar diffusion calculation of §12.6. This was for the mass diffusion between an infinite reservoir of argon at a number density of 2.8×10^{20} m^{-3} and a temperature of 273 K at $x=0$ and a similar reservoir of helium at $x=1$ m. The concentration gradients and the resulting diffusion velocities were discussed in §12.6, and the temperature and number density in the diffusing flow did not depart from the reservoir values. However, the stream velocity and heat flux did exhibit systematic effects.

The stream velocity component is in the direction of the argon diffusion and ranges from 2 to near 20 ms^{-1}. However, as shown in Fig. 12.17, the product of this velocity with the density is uniform across the flow and represents a net mass flux in the positive direction.

Fig. 12.18 shows that the net heat flux q_x is in the negative direction and ranges from 20 to 66 Wm^{-2}. Chapman and Cowling (1970) show that, in the absence of a temperature gradient,

$$q_x = (5/2)kT(n_1U_1+n_2U_2) + kT(n_1n_2/n)\alpha_T(U_1-U_2). \qquad (12.22)$$

The first term appears because the flux is measured relative to the stream or mass velocity of the gas, rather than the mean velocity of the molecules. This is also plotted in Fig. 12.18 and is in the same direction as q_x, but is slightly larger in magnitude. The small difference is the diffusion thermo-effect that is represented by the third term in eqn (12.22). This is in the direction of the argon diffusion and, as noted by Chapman and Cowling 'the heat flow is in the direction of diffusion of molecules which, in thermal diffusion, would seek the cold side'. Because the effect is the small difference between large quantities the scatter is high, but the average value of α_T across the flow is 0.44, and this is in reasonable agreement with the measured value of 0.38.

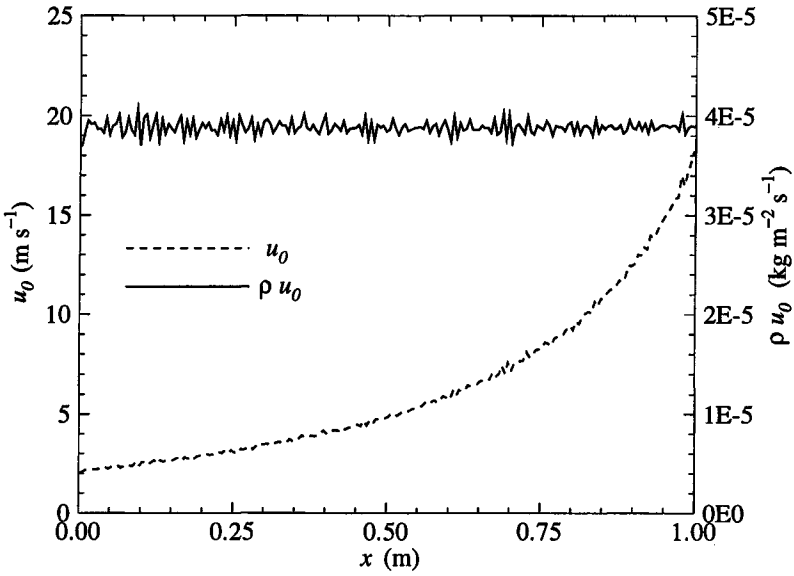


Fig. 12.17 The stream velocity and mass flux in the He–Ar diffusion.

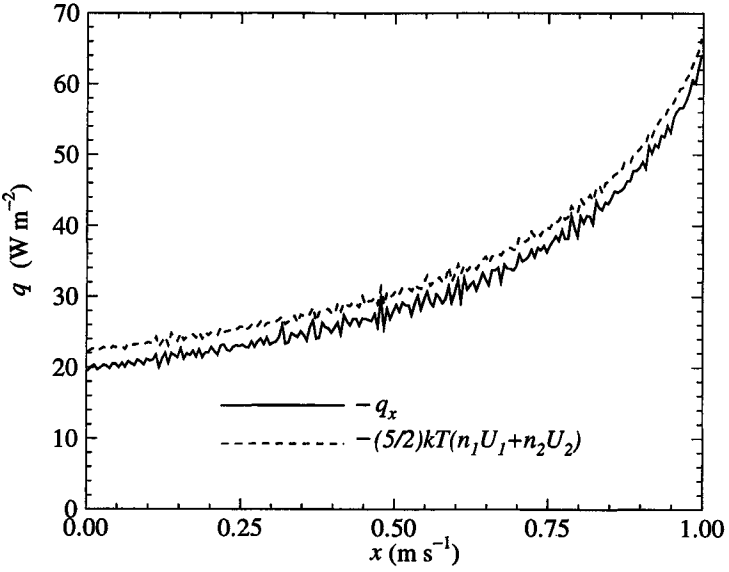


Fig. 12.18 Heat fluxes in the He–Ar diffusion flow.

12.9 Heat transfer in the transition regime

The DSMC calculations in the preceding sections of this chapter were made for sufficiently low Knudsen numbers that the rarefaction effects were confined to the Knudsen layers near the boundaries, and the Chapman–Enskog transport properties were applicable to the rest of the flow. The one-dimensional heat transfer between plane surfaces will now be studied over the full transition regime from continuum to free molecule flow. The free-molecule or collisionless solution was presented, together with the continuum solution, in §7.3. The four-moment solution of Liu and Lees (1961) for this problem in the transition regime was presented in §8.2.

The four-moment solution was for the special case of Maxwell molecules for which the coefficient of heat transfer $K = CT$. Eqns (3.56) and (4.62) show that this relationship can be reproduced by the VHS model if C is given by

$$C = \frac{15k^{3/2}}{8(\pi m T_{\text{ref}})^{1/2} d_{\text{ref}}^2}. \tag{12.23}$$

The comparison between the DSMC and analytical results for the heat flux is most conveniently made if the pressure p on the surfaces is chosen as the independent variable. This pressure in the free-molecule flow is readily shown to be

$$p = nk T_{\text{U}}^{1/2} T_{\text{L}}^{1/2}, \tag{12.24}$$

where T_{U} and T_{L} are the temperatures of the upper and lower plates, respectively. Note that the number density n becomes a constant in the collisionless limit. The free molecule solution of eqn (7.24) can then be written

$$q_{\text{x}} = -2^{3/2} p \left(\frac{k}{\pi m}\right)^{1/2} (T_{\text{U}}^{1/2} - T_{\text{L}}^{1/2}). \tag{12.25}$$

The continuum heat flux is independent of the pressure and, since $\omega = 1$, it can be written from eqn (7.25) as

$$q_{\text{c}} = -\tfrac{1}{2} C (T_{\text{U}}^2 - T_{\text{L}}^2)/h, \tag{12.26}$$

where h is the distance between the surfaces. The four-moment solution is, from eqns (8.19) and (8.20),

$$q = \frac{\{(BT_{\text{U}} + AT_{\text{L}} + h/C)^2 - (B^2 - A^2)(T_{\text{U}}^2 - T_{\text{L}}^2)\}^{1/2} - (BT_{\text{U}} + AT_{\text{L}} + h/C)}{B^2 - A^2},$$

where (12.27)

$$A = \frac{1}{2p}\left(\frac{\pi m T_{\text{U}}}{2k}\right)^{1/2} \quad \text{and} \quad B = \frac{1}{2p}\left(\frac{\pi m T_{\text{L}}}{2k}\right)^{1/2}.$$

This relation reduces to the continuum result when the pressure becomes large, and to the free-molecule value at very small pressures.

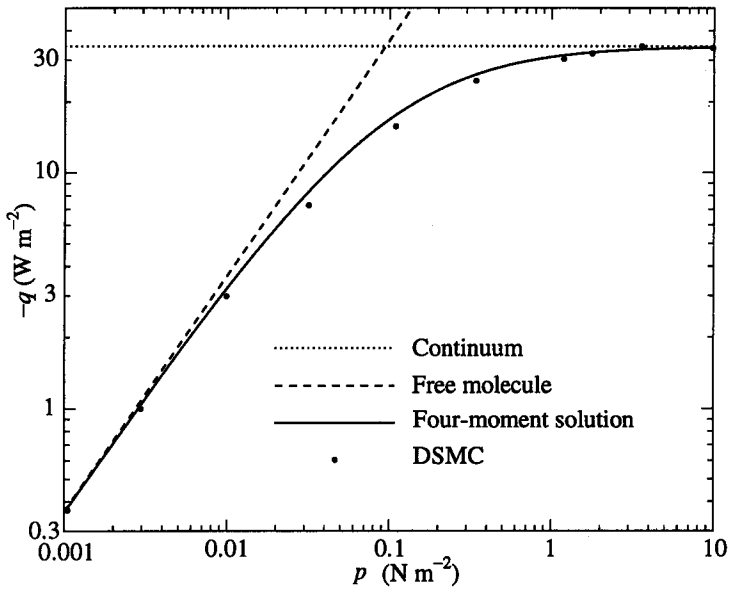


Fig. 12.19 Heat transfer in the transition regime.

Program DSMC1 was applied to this problem with a surface at 250 K as the inner boundary and a surface at 1000 K as the outer boundary. The VHS molecular model was employed with the molecular mass and diameter appropriate to argon, but with $\omega = 1$. The results are compared with the continuum, free molecule, and four-moment solutions in Fig. 12.19. This shows that the four-moment solution is remarkably accurate in predicting the pressure range over which the transition takes place, and the heat transfer values from this solution are about 10% too high over most of the transition region. It might be thought that this difference may be due to the Maxwell collision integral on which the four-moment solution depends not being consistent with the VHS model. However, an earlier DSMC study of this problem (Bird 1976) used the Maxwell molecular model and showed a similar difference. Moreover, this study showed that there is little difference between the results from hard sphere and Maxwell molecules for this problem.

The Knudsen number range may be inferred from a comparison of Fig. 12.19 with Fig. 8.1. This latter figure is misleading in that it suggests that the heat transfer increases with Knudsen number. This is due to the heat transfer being expressed as the ratio to the free-molecule value and, as shown Fig. 12.19, the absolute value decreases with increasing Knudsen number. There is some scatter in the DSMC results at low Knudsen numbers because q_x is then the small difference between large quantities. The scatter at high Knudsen numbers is less by orders of magnitude.

12.10 Breakdown of continuum flow in expansions

A steady isentropic expansion is one of the basic flows of gas dynamics. Exact solutions of the continuum equations are available but, because the flow is isentropic, these do not include any of the transport properties. Therefore, the breakdown in the continuum solution at sufficiently low densities cannot be associated with the breakdown of the Chapman–Enskog theory. It occurs simply because the collision frequency drops below the rate that is required for maintenance of the continual drop in temperature that is predicted by the theory. This form of breakdown was discussed in §1.2, and it was suggested that its onset in a steady flow should correlate with the parameter P that was defined in eqn (1.5) as

$$P = (\pi^{1/2}/2)\, s\, (\lambda/\rho)\, \mathrm{d}\rho/\mathrm{d}x\,.$$

Program DSMC1 has been used to study this form of breakdown in steady spherical and cylindrical expansions. Because the area of these flows increases monotonically with radius, only the supersonic branch of the expansion can be modelled. An overall Knudsen number is most conveniently defined as the ratio of the mean free path under stagnation conditions to the radius at which the flow is sonic, i.e.

$$(Kn) = \lambda_0/r^*\,, \tag{12.28}$$

where the subscript $_0$ denotes stagnation conditions and the superscript * denotes sonic conditions. For a flow in which the area is proportional to the distance to the power ε, the parameter P can then be written (Bird 1970)

$$P = \frac{(Kn)}{2}\left(\frac{\pi\gamma}{2}\right)^{1/2} \frac{\varepsilon\,(Ma)^{3+1/\varepsilon}}{(Ma)^2-1}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2\varepsilon(\gamma-1)}} \times\left(1+\frac{\gamma-1}{2}(Ma)^2\right)^{\frac{\gamma+1}{2(\gamma-1)}\left(1-\frac{1}{\varepsilon}\right)-\omega}. \tag{12.29}$$

The singular behaviour of P at the sonic line reflects the fact that the cylindrical and spherical geometries do not satisfy the requirement that $\mathrm{d}A/\mathrm{d}x$, and therefore ε, should be zero at (Ma)=1. P drops to a minimum at a Mach number of the order of 1.4 and then increases. It was found that, if the computation was started at the sonic line, the initial axial and radial temperatures were slightly out of equilibrium. While they rapidly came back into equilibrium, most calculations avoided the singularity by starting the computation at a radius equal to 1.05 times the sonic radius. For a spherical flow, this radius corresponded to a Mach number of just over 1.4.

The first calculation was for the spherical expansion of argon with the overall Knudsen number, as defined in eqn (12.28), equal to 0.001. The flow extended from x=1.05 to 10. There were 985 cells, 9,850 sub-cells and approximately 35,000 molecules in the steady state. The inner boundary was a stream that matched the continuum solution, while the outer boundary was a vacuum. The initial state of the flow was also a vacuum.

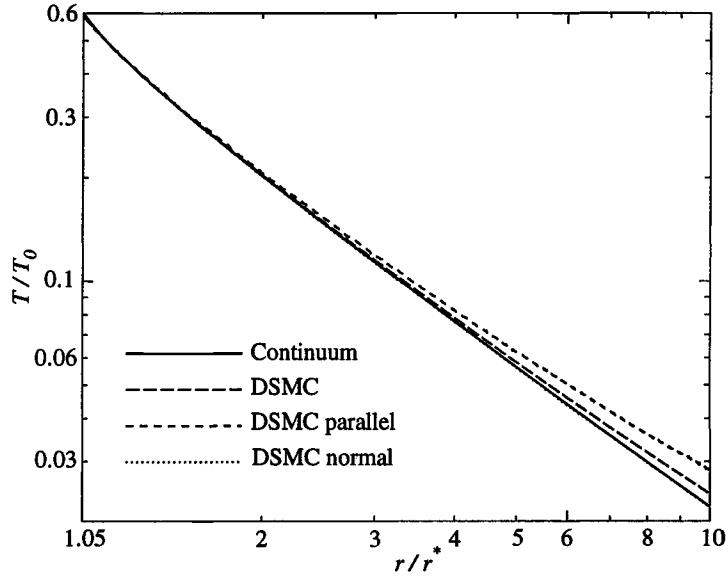


Fig. 12.20 Temperature ratio in a $(Kn) = 0.001$ expansion of argon.

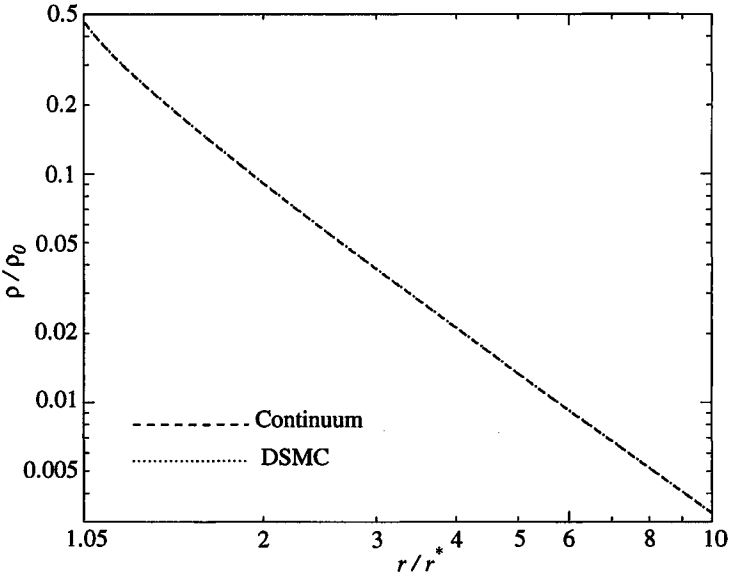


Fig. 12.21 Density ratio in a $(Kn) = 0.001$ expansion of argon.

The total sample size in this calculation was almost one thousand times larger than that in earlier calculations of this type (Bird 1970), and the statistical scatter was generally a small fraction of one percent. The temperature and density profiles in the expansion are compared with the continuum profiles in Figs 12.20 and 12.21, respectively. The earlier calculations were for hard sphere and Maxwell molecules, and argon with $\omega = 0.81$ is nearer to the Maxwell case. The breakdown shows as a very gradual rise of the temperature above the continuum value. This is associated with a separation of the parallel or axial temperature T_x based on the thermal velocity components in the x-direction and the normal or circumferential temperature T_n that is based on the components normal to this direction. This latter temperature is the mean of the y- and z-components of temperature that are output, along with the x-component, in the output on the 'separate species'. The parallel temperature would eventually become almost constant or 'frozen'. On the other hand, the normal temperature continues to fall, even in a collisionless flow, because of the geometric effect that causes the molecular velocities to become more nearly radial as they move further from the centre. In this case, the normal temperature remains very close to the continuum temperature. The ratio T_x/T_n provides the most convenient measure of the extent of the non-equilibrium, and this is plotted against P in Fig. 12.22.

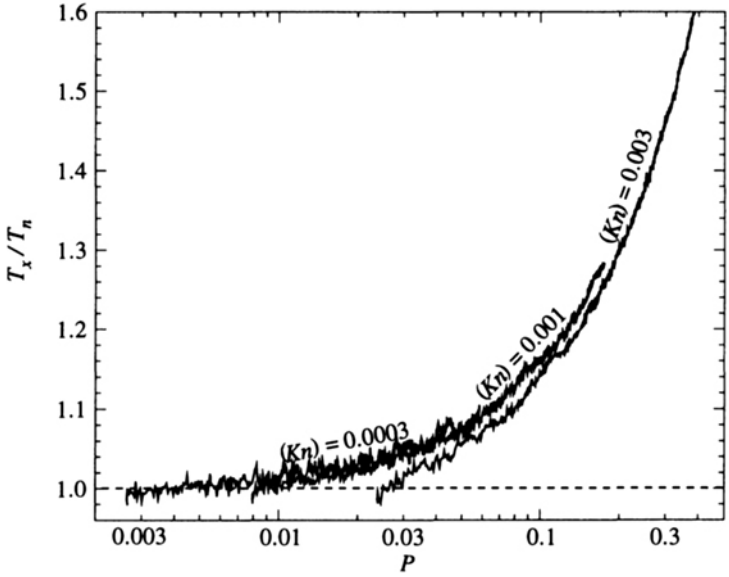


Fig. 12.22 Ratio of the parallel and normal temperatures as a function of the breakdown parameter.

The calculation was repeated for a density one third that of the original case and also for a density three times that of the original case. These are referred to as the $(Kn) = 0.003$ and the (Kn)=0.0003 cases, respectively. The qualitative behaviour was the same for all three cases, and the T_x/T_n ratios are also shown in Fig. 12.22. The results for the (Kn)=0.001 case range from $P = 0.008$ to 0.16. The breakdown for this case commences near $P = 0.01$, which occurs at $r/r^* = 1.28$ where the Mach number is 2.1. The results for the $(Kn) = 0.0003$ case range from $P = 0.0025$ to 0.045, and the breakdown again commences near $P = 0.01$ which occurs here at a Mach number of 4.35. The $(Kn) = 0.003$ case commences at $P = 0.025$ and, for this value, the flow should be out of equilibrium. The boundary condition forces it to be initially in equilibrium and this produces an overall downward displacement of the curve.

The results give very strong support for the use of the parameter P to correlate the breakdown of equilibrium in expansions. The results are in agreement with the early DSMC results (Bird 1970) that led to a proposed breakdown criterion of $P = 0.04$. This comparatively large value was due to the high level of scatter in the original calculations, and the fact that most of these started (like the $(Kn) = 0.003$ case here) at a radius where the flow should already have been out of equilibrium. The temperature ratio at $P = 0.04$ is approximately 1.06, and Fig. 12.20 indicates that the overall temperature is then 2% above the theoretical temperature. In view of the fact that the density and velocity are not affected, this could well remain a suitable criterion, even though the current results indicate that the breakdown starts near P=0.01. Also, the breakdown is so gradual that, if the scatter was reduced further in much larger calculations, some breakdown would be observed at even lower values of P. It is worth noting that the fractional departure of the temperature above the continuum value is approximately equal to $P/2$.

It has sometimes been suggested that the breakdown of equilibrium in a high mass-ratio mixture would lead to pronounced separation effects. The $(Kn) = 0.001$ case was therefore repeated for a mixture of equal parts by mass argon and helium. The results for the temperature, density, and temperature ratio T_x/T_n as a function of the breakdown parameter P were indistinguishable from the results for argon as a simple gas. However, there was a diffusion velocity of approximately -5 ms^{-1} for the argon and a similar positive diffusion velocity for the helium. These are very small in comparison with the final stream velocity of almost 1300 ms^{-1}, and the separation effect was small. The final argon number fraction was found to be about 0.1005, compared with the value of 0.1 that would be obtained in the absence of separation. Moreover, this should not be regarded as a 'breakdown' effect because the diffusion velocities are consistent with the predictions of the diffusion equation (3.91) that is based on the Chapman–Enskog transport properties. The pressure and thermal diffusion terms are much larger than the mass diffusion term. These two effects act in opposite directions in a steady expansion, but the pressure diffusion term is the dominant one.